

#8.6

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

a. $N=1000$ $SSE_M = 97161.9174$ $\hat{\sigma}_F^2 = 12.024$ $\hat{\sigma}_M^2 = \frac{97161.9174}{577-4} = 169.567$

① $\begin{cases} H_0: \sigma_M^2 = \sigma_F^2 \\ H_a: \sigma_M^2 \neq \sigma_F^2 \end{cases}$ ② $\alpha = 0.025$ ③ $F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} \stackrel{H_0}{\sim} F(577-4, 423-4)$

④ $RR = \{F \geq F_{0.025}(419, 573) = 1.1968\}$ ⑤ $F_0 = \frac{169.567}{12.024} \approx 1.4129$

⑥ $\because F_0 \notin RR$, do not reject H_0

b. ① $\begin{cases} H_0: \sigma_M^2 = \sigma_S^2 \\ H_a: \sigma_M^2 > \sigma_S^2 \end{cases}$ ② $\alpha = 0.05$ ③ $F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_S^2} \stackrel{H_0}{\sim} F(600-5, 400-5)$

④ $RR = \{F \geq F_{0.05}(595, 395) = 1.1647\}$ ⑤ $F_0 = \frac{100,703.0471/595}{56,231.0382/395} = 1.1869$

⑥ $\because F_0 \in RR$, reject H_0

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

C. ① $\begin{cases} H_0: \alpha_2 = \alpha_3 = \dots = \alpha_5 \\ H_A: \text{not all the } \alpha_i \text{ in } H_0 \text{ are zero} \end{cases}$ ② $\alpha = 0.05$

③ $\chi^2 = NR^2 \sim \chi^2(5-1)$ ④ $RR = \{ \chi^2 > \chi^2_{0.05}(4) = 9.488 \}$

⑤ $NR_0 = 59.03$ ⑥ $\because NR_0 \in RR$, 故 reject H_0

⇒ 此結果顯示異質 Var 的存在支持 (b) 的猜測, wage 的變異可能會隨著其他特徵 (eg 婚姻) 而變動, 但無法確定是否在不同婚姻狀況有差異, 因為 married 不是 Model 中的變數。

D. $df = 4 + 2 + 6 = 12$

交乘項 = $4 \times 3 + 2 = 6$

④ $RR = \{ \chi^2 > \chi^2_{0.05}(12) = 21.03 \}$

⑤ $LM_0 = 78.82$

⑥ $\because LM_0 \in RR$, 故 reject H_0 , Model 存在異質變異數。

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

	se	robse	
intercept:	2.36	< 2.5	) wider
EDUC :	0.14	< 0.16	
EXPER :	0.031	> 0.029	) narrower
METRO :	1.05	> 0.84	
FEMALE :	0.81	> 0.8	

不存在inconsistency。在存在異質變異數的情況下，OLS 計算出的標準誤是有偏誤的，它們可能低估也可能高估真實的標準誤差，而穩健標準誤可以修正這個偏差。因此，有些係數的標準誤上升、有些下降是非常正常的現象。

f.

在 (f) 中發現married的t值約為 1.0（不顯著），而 (b) 小題則證實了已婚與否會導致明顯的異質變異數，這兩者是相容的（Compatible），並不衝突。

- (f) 的結果：t值約 1.0 代表婚姻對工資的平均水準沒有顯著影響。平均而言，已婚者與單身者的薪水沒有統計上的顯著差異。
- (b) 的結果：證明了婚姻狀況會影響工資的變異數。雖然已婚者平均賺得沒有比較多，但他們的工資「波動範圍」或「不確定性」與單身者顯著不同（通常更大）。

#8.16(a)

95% CI for KIDS (OLS)

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	Variable	Lower_95	Upper_95

1	kids	-135.330	-28.323

Holding income and age constant, the 95% confidence interval for the effect of one additional child on miles traveled is **-135.33 ~ -28.323 miles**.

#8.16(b)



The "Residuals vs INCOME" plot exhibits a funnel shape (the spread of residuals increases as income increases). This pattern strongly suggests the presence of **heteroskedasticity**.

#8.16(c)

Goldfeld-Quandt Test Results		
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Test	GQ_Statistic	p_value

1 Goldfeld-Quandt	3.104	0.00000

H0: Homoskedasticity (Error variances are equal).

H1: Heteroskedasticity (Error variance increases with income).

Conclusion: The GQ test yields a p-value of **0.00000**. Because the p-value is less than 0.05, we reject the null hypothesis and conclude that heteroskedasticity is present at the 5% significance level.

#8.16(d)

Comparison of OLS Confidence Intervals		
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Model	Lower_95	Upper_95

OLS Default CI	-135.330	-28.323
kids OLS Robust CI	-139.323	-24.330

#8.16(e)

GLS vs OLS Confidence Intervals		
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Model	Lower_95	Upper_95

1 OLS Robust (from d)	-139.323	-24.330
2 GLS Conventional	-119.894	-33.718
3 GLS Robust	-121.413	-32.199
